Braeden Burger

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https://github.com/2002goblue/

Objective

Motivated Computer Science student in the BS/MS program at the University of Michigan–Flint with strong skills in C++, Python, Java, and C#. Seeking a **Software Development Internship** to contribute to the design, implementation, and testing of reliable software solutions. Experienced in applying principles of **software engineering**, **AI integration**, and **human-computer interaction** to create efficient, user-focused applications. Proven ability to collaborate in multidisciplinary teams, debug complex systems, and deliver high-quality results through iterative development and data-driven problem solving.

Education

University of Michigan-Flint | Flint, MI
Bachelor of Science in Computer Science (4.0 GPA) Expected Apr. 2026
Master of Science in Computer Science and Information Systems (SWE Concentration) Expected Apr. 2027

Work Experience

University of Michigan-Flint | Flint, Michigan Aug. 2024 - Present
Research Assistant

- Implementing and debugging applications in **C#** and **Python**, integrating **AI** and **networking components.**
- Work collaboratively with a multidisciplinary team to design, implement, and evaluate complex software systems.
- Conducting qualitative and speculative HCI research on digital consent, cross-reality interaction, and social VR.
- **First author of a full paper** that was accepted and presented at the 2025 ACM CHI conference in Tokyo, Japan.
- **Co-author and presenter of a full paper** accepted to the 2025 ACM CSCW conference in Bergen, Norway.

National Science Foundation (REU) | Flint, Michigan May 2025 - Aug. 2025
NSF REU Researcher (Internship)

- Designed and built an **XR interaction application** in **C#** for smart glasses using real-time data processing.
- Integrating **generative and multimodal AI** in XR glasses, enabling context-aware interactions in social settings.
- Developed **in-depth user studies** aimed at informing future approaches to safer and more intuitive XR design.
- Managed project timelines and documentation while iterating on designs through usability testing and feedback.

National Science Foundation (REU) | Flint, Michigan May 2024 - Aug. 2024
NSF REU Researcher (Internship)

- **Managed a team of researchers** in the development of an innovative **Bluetooth-connected haptic device.**
- Used **C++** and **embedded firmware** to enhance communication of individuals with hearing and visual disabilities.
- Analyzed user testing data to evaluate haptic device performance and guide **iterative design improvements.**

University of Michigan-Flint | Flint, Michigan Oct. 2023 - Sep. 2024
Undergraduate Researcher (UROP)

- Analyzed and categorized speculative qualitative data to investigate emerging interaction concepts in social VR.
- **Co-authored a CHI workshop paper** and participated in the **ACM CHI 2024** workshop in Honolulu, Hawaii.
- **Co-authored a short paper accepted to CSCW 2024** exploring consent and safety in cross-reality spaces.
- Presented findings at UM MISC Research Showcase and the College of Innovation & Technology Project Day.

Technical Skills

- **Languages:** C++, Python, Java, C#, JavaScript, HTML/CSS
- **Frameworks/Tools:** VS Code, GitHub, Linux, Node.js, MongoDB, Arduino
- **Concepts:** Data Structures and Algorithms, Computer Networking, XR UI/UX
- **Research & Design:** Human-Computer Interaction, User-Centered Design, Data Analysis, Qualitative Research

Projects

University of Michigan-Flint | Flint, MI
DrillVision - AI-Powered Marching Band Drill Analyzer Aug. 2024 - May 2025

- Developed with **Python, PyTorch, OpenCV, Unity, Node.js, AWS**
- **Developed a computer-vision system** to detect and analyze marching band formations using Faster R-CNN.
- Implemented IoU/mAP metrics for validation and **trained models with custom datasets of band images.**
- Deployed backend on **AWS EC2** using **Node.js** for model inferencing, data processing, and API handling.
- Configured **AWS S3** for secure storage and retrieval of images, datasets, and processed model outputs.